6COSC020W Applied AI

Tutorial Week 7: Machine Learning

Aim**:**

- Review machine learning concepts.
- Experiment with TensorFlow and machine learning.
- Train and evaluate machine learning models for classification and regression.
- Explore new problems and datasets.

1.- Definitions and concepts:

Question 1) What are the three different types of machine learning? List and describe them using your own words.

Question 2) You have been asked to implement a system that tells us the expected final grade of our students based on their results from level 4 and level 5. Which algorithm will you use? Explain why you chose your preferred algorithm and which data you would use.

Question 3) You have been asked to implement a system that, given information about customer purchases, you need to classify them in unknown buying behaviours or preferences. Which algorithm will you use?

Question 4) Explain the K-means algorithm using your own words.

2.- Tutorial_Week7.ipynb.

1) Download the file *Tutorial_Week7.ipynb* from Blackboard.
2) Upload and open the file with Google Collaboratory (recommended) or Anaconda (selecting Jupyter Notebook).
3) Work through the Jupyter Notebook by answering the questions and doing the tasks.